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LLM-based Agents Suffer from Hallucinations: A Survey of Taxonomy, Methods, and Directions

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***Abstract*—Driven by the rapid advancements of Large Language Models (LLMs), LLM-based agents have emerged as powerful intelligent systems capable of human-like cognition, reasoning, and interaction. These agents are increasingly being deployed across diverse real-world applications, including student education, scientific research, and financial analysis. However, despite their remarkable potential, LLM-based agents remain vulnerable to hallucination issues, which can result in erroneous task execution and undermine the reliability of the overall system design. Addressing this critical challenge requires a deep understanding and a systematic consolidation of recent advances on LLM-based agents. To this end, we present the first comprehensive survey of hallucinations in LLM-based agents. By carefully analyzing the complete workflow of agents, we propose a new taxonomy that identifies different types of agent hallucinations occurring at different stages. Furthermore, we conduct an in-depth examination of eighteen triggering causes underlying the emergence of agent hallucinations. Through a detailed review of a large number of existing studies, we summarize approaches for hallucination mitigation and detection, and highlight promising directions for future research. We hope this survey will inspire further efforts toward addressing hallucinations in LLM-based agents, ultimately contributing to the development of more robust and reliable agent systems.**

***Index Terms*—LLM-based Agents, Hallucinations, Safety.**

I. Introduction

Language Models (LLMs) [1], [2], [3], [4], [5] have recently showcased extraordinary capabilities across a broad spectrum of tasks, including language generation [6], intent comprehension [7], and knowledge reasoning [8]. These capabilities are largely attributed to the vast scale of training data [9], model architecture innovations [10], and emergent abilities [11] that arise during instruction tuning [12] and in-context learning [13]. Building on these breakthroughs, **LLM-based Agents** [14], [15], [16], [17], [18] are becoming increasingly proficient in task automation across a wide range of fields, marking a critical milestone on the path toward Artificial General Intelligence (AGI) [19]. Moreover, these agents can be orchestrated into a **LLM-based Multi-agent System** (MAS) [20], [21], [22], [23], [24], where different agents with distinct specializations collaborate and interact to solve real-world and complex problems beyond the capacity of any single agent through mutual cooperation, such as knowledge sharing [25] and collaborative coordination [26].

Despite the impressive performance achieved by LLM-based agents, their rapid advancement introduces a spectrum of safety challenges [27], [28], [29]. Among these challenges, **Agent Hallucinations** represents a particularly significant threat [30], [31], [32]. Previous studies on hallucinations have primarily focused on the field of natural language generation (NLG) [13], where hallucinations refer to the phenomenon in which NLG models generate unfaithful or nonsensical text. Ji et al. review recent advances in addressing hallucinations across various NLG tasks [33]. Meanwhile, Huang et al. specifically focus on the causes of LLM hallucinations [34]. In this context, hallucinations in LLMs are categorized into factuality and faithfulness hallucinations. Factuality hallucinations highlight discrepancies between the generated content and verifiable real-world facts [35]; while faithfulness hallucinations refer to deviations from the user's original input [36]. In addition, the authors discuss several effective methods for detecting and mitigating hallucination issues [37].

Unlike the aforementioned settings, the LLM-based agent is a more sophisticated intelligent system, equipped with goal-directed reasoning and action-taking capabilities. Such agents typically comprise three core modules: brain, perception, and action [38]. The brain module is primarily responsible for storing memory and knowledge, supporting reasoning and decision-making for executing tasks; the perception module extends the agent's perceptual space, enabling it to handle multi-modal environmental inputs; and the action module expands the agent's action space, allowing it not only to generate

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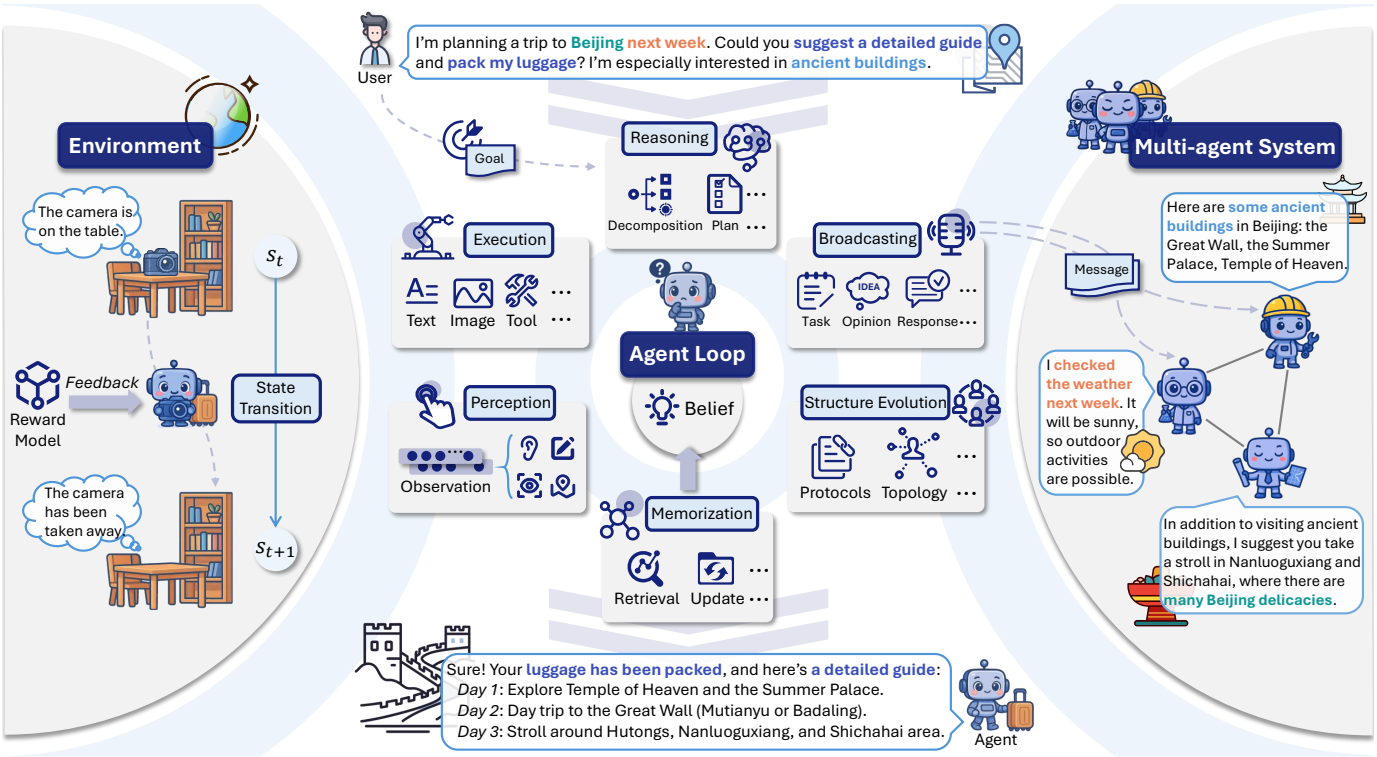

Fig. 1. An overview of agent goal completion. Within the loop, the LLM-based agent carries out external behaviors such as reasoning, execution, perception, and memorization, guided by its internal belief state. Throughout this process, the environment dynamically evolves in response to the agent's decisions, while task allocation within the LLM-based multi-agent system including broadcasting and structure evolution further enhances the fulfillment of user requirements.

textual outputs but also to invoke tools for completing more complex tasks. Therefore, in LLM-based agents, hallucinations are not "linguistic errors", but rather a broad category of fabricated or misjudged "human-like behaviors" that may occur at any stage of the agent's pipeline. Accordingly, the manifestations and causes of agent hallucinations are considerably more complex. This complexity is reflected in three key aspects: **1) More Diverse Types**: Rather than the straightforward response errors of a single model, agent hallucinations are compound behaviors arising from interactions among multiple modules, resulting in a broader and more varied range of hallucination types. **2) Longer Propagation Chains**: The aforementioned hallucinations are mostly localized and single-step errors, whereas agent hallucinations often span multiple steps and involve multi-state transitions. Such hallucinations are not limited to the final output; they may also arise during intermediate processes such as perception and reasoning, where they can propagate and accumulate over time. **3) More Severe Consequences**: Agent hallucinations involve "physically consequential" errors, where incorrect embodied actions can directly affect task execution, system devices, and user experiences in the real world. As a result, the cost and risk associated with agent hallucinations are significantly higher. However, existing reviews of LLM-based agents primarily focus on architectural designs and practical applications, while giving far insufficient attention to the importance and urgency of agent hallucinations.

To this end, we provide an overview of agent hallucinations to fill this important gap and promote the advancement of agents. In this paper, the interaction dynamics of LLM-based agents are formulated as a partially observable Markov decision process (POMDP) where the agent[1] interacts with the learning environment, makes decisions, receives feedback, and updates its state over multiple time steps. Based on this general POMDP setting, we summarize the contributions of our work below.

- **First Survey.** To the best of our knowledge, this is the first survey to review hallucination issues in LLM-based agents. It encompasses recent research on both mitigation and detection methods, offering a broad perspective on the development of LLM-based agents.
- **Innovative Taxonomy.** We introduce a novel decomposition of agent components into two parts: internal state and external behaviors. The former is characterized by a belief state maintained by the agent, which serves as the most fundamental unit of the agent cognition policy. External behaviors refer to a series of proactive procedures guided by the belief state. Through this internal–external distinction, we can classify agent hallucinations based on the specific stage at which they occur, covering five types of agent hallucinations.
- **Comprehensive Review.** For each type of agent hallucinations, we offer a formal definition, illustrative examples, and an in-depth discussion of representative studies. Based on this, we identify eighteen triggering causes of agent hallucinations. Furthermore, we summarize ten

[1]Throughout this paper, we use the term "agent" to denote the LLM-based agent, unless otherwise specified.

general approaches for hallucination mitigation, along with corresponding detection methods, to provide readers with a clear and up-to-date overview of the current research landscape on agent hallucinations.
- **Future Outlook.** By reviewing and summarizing existing solutions to hallucination problems and their possible limitations, we outline several promising directions for future exploration that will need to be fully investigated to advance both academic research and real-world deployment of LLM-based agents.
- **Open Resource.** We provide a well-curated collection of resources, encompassing over 200+ related papers, which we make publicly available on GitHub[2] to foster community engagement and collaboration.

Paper Organization. The remainder of this paper is structured as follows. Section II introduces the formal definition of LLM-based agents. Based on this definition, Section III presents a new taxonomy of agent hallucinations. Section IV reviews existing solutions, with particular emphasis on methodologies for hallucination mitigation. Section V outlines future research directions. Section VI concludes the survey.

II. Formal Definition of LLM-based Agents

Fig. 1 provides an overview of agent goal completion. The formal definition of LLM-based agents including interaction dynamics and LLM-based agent loop is given here.

A. Interaction Dynamics

The interaction dynamics between the agent and the learning environment is usually formulated as a Partially Observable Markov Decision Process (POMDP) [39] which is defined as a 8-tuple: $\mathcal{E} = (\mathcal{S}, \mathcal{A}, T, \mathcal{G}, \mathcal{O}, Z, R, \gamma)$. The description of each element in $\mathcal{E}$ is given here:

- $\mathcal{S}$ is the state space where $s \in \mathcal{S}$ represents a true environment state. A POMDP explicitly models an agent decision process in which the agent cannot directly observe the underlying state. Instead, the agents must maintain a **Belief State** to represent its subjective understanding of the learning environment.
- $\mathcal{A}$ is the action space. An action $a \in \mathcal{A}$ involves content generation or the use of external tools, such as scheduling events via calendar APIs and issuing smart home control commands.
- $T : \mathcal{S} \times \mathcal{A} \to P(\mathcal{S})$ is the state transition probability function. For each state-action pair (s, a), it specifies a probability distribution over possible subsequent states. For example, executing a "turn off lights" action may result in reaching the desired state with 90% probability, while network latency or system errors could cause the state to remain unchanged with 10% probability.
- $\mathcal{G}$ is the goal space. Each goal $g \in \mathcal{G}$ specifies a particular objective of users.
- $\mathcal{O}$ is the observation space. Each observation $o \in \mathcal{O}$ represents a partial view of s, which can be manifested through different modalities. Partial observability may stem from multiple factors, such as the complexity of the learning environment and the limitations of the agent's perceptual capabilities.
- $Z : \mathcal{S} \times \mathcal{A} \to \mathcal{O}$ is the observation function. For each state-action pair (s, a), it provides a possible observation o for the agent.
- $R : \mathcal{S} \times \mathcal{A} \to \mathcal{R}$ is the reward function, where $\mathcal{R}$ is the reward space. For each state-action pair (s, a), it provides a feedback $r \in \mathcal{R}$, and r is usually expressed in numerical form.
- $\gamma \in [0, 1)$ is the discount factor. It serves to balance the importance between immediate and future rewards. Since events that occur farther in the future are harder to predict accurately, it is generally desirable to reduce the significance of long-term rewards.

B. LLM-based Agent Loop

Given a specific goal g, accomplishing it typically requires multiple execution loops. Furthermore, since the true environment state cannot be directly observed, an LLM-based agent needs to maintain a belief state b_t that represents its internal and subjective understanding of the learning environment. In the multi-loop process, the agent continuously updates b_t, allowing b_t to be dynamically refined across diverse contexts and over extended time spans. b_t is the most fundamental unit in the agent's loop process, forming the basis of all agentic operations. Based on b_t, we further introduce the agent's external behaviors including reasoning, execution, perception, and memorization. The concrete calculation in each loop can be given here:

- **Reasoning**: The agent first generates a plan p_t for the next action conditioned on b_t and g:
$$p_t = \Phi(b_t, g). \tag{1}$$
- **Execution**: The agent then translates p_t into an executable action a_t:
$$a_t = E(b_t, p_t). \tag{2}$$
a_t typically involves the invocation of relevant external tools.
- **Feedback**: The learning environment would objectively provide a reward r_t based on s_t and a_t:
$$r_t = R(s_t, a_t). \tag{3}$$
- **Environment Transition**: The learning environment transitions to the next state $s_{t+1} \in \mathcal{S}$ according to the following generated probability distribution:
$$\Pr(s_{t+1}|s_t, a_t) = T(s_t, a_t). \tag{4}$$
- **Perception**: The agent would then perceive s_{t+1} and generate the observation $o_{t+1} \in \mathcal{O}$:
$$o_{t+1} = Z(s_{t+1}, b_t, a_t). \tag{5}$$
- **Memorization**: Both a_t and o_{t+1} are used for updating its external memory module:
$$m_{t+1} = L_M(m_t, a_t, o_{t+1}). \tag{6}$$

[2]https://github.com/ASCII-LAB/Awesome-Agent-Hallucinations.